

Computer and Information Science Department

CS208 Individual Coursework Assignment 2

Weighting: 7.5% of Overall CS208 Mark

Marks Available: 10

Important Note: Gen-AI is not to be used for the completion of this assignment.

Goal

In this coursework, you will be tasked with solving a multi-processor scheduling problem. You have some processors that can execute each of the jobs, and each processor will complete the jobs in a certain amount of time. Processors are well suited to specific jobs and will complete those jobs more quickly. You are tasked with building an algorithm that can provide an optimal solution to each of the problems.

The Problem

You are given the following jobs to schedule on a processor. For each, you must develop an algorithm to solve the problem. There are 16 processors to which jobs can be assigned and 512 jobs in total, with 12 different problem sets. Your overall goal is to minimize the makespan, which is the total amount of time that is required for **all** jobs to be completed. Another way of describing the makespan is that it is the time taken for the processor with the most amount of work to finish.

The files provided give you the run time for 512 jobs on each of the 16 different computers. The job changes every 16 lines, and each line refers to the runtime of that job on the corresponding processor. Here are some examples:

Line 1 is the runtime of job 1 on computer 1, line 2 is the runtime of job 1 on computer 2. Line 17 is the runtime of job 2 on computer 1, line 18 is the runtime of job 2 on computer 2 etc.

Part 1: Min-Min Algorithm

- Implement the Min-Min algorithm and run it on the 12 instances

Part 2: Local Search

- Implement the Local Search algorithm and run it on the 12 instances
 - You have a maximum of 4.5 minutes to run your local search.

Part 3: Make Improvements

- You have two algorithms which can solve the problem. Is there anything else that you can do with them to achieve better results?

Deliverables:

- The source code that you have developed.

General Instructions

Spend some time looking at the example provided in the code, RoundRobinAlgorithm. This will give you some hints and ideas for how to build a solution to the problem. You need to construct a list of the total amount of time allocated to each processor.

The class named “YourAlgorithm” is where you should put your code. Rename it to FirstNameLastNameAlgorithm and change the name field in the class to your own name.

You will notice that there is a class named “AlgorithmEvaluator”. There is a method here named “evaluateAlgorithm” which can be used to see what grade you might get for your submission with a given file. Use this to help you figure out if you are on the right path with your submission.

Each file is read in and converted into what is known as an Expected Time to Compute Matrix (ETC). This is a matrix that specifies how long each job will take to complete on each processor.

For example, if the first table is the input for two processors:

Job1	10
Job1	15
Job2	4
Job2	6
Job3	42
Job3	64

Then the ETC matrix will look like this:

Job	Processor1	Processor2
Job1	10	15
Job2	4	6
Job3	42	64

General Advice

1. Understand the Problem Clearly

Your goal is to minimize the makespan—the total time required to complete all jobs.

You have 16 processors and 512 jobs, with varying execution times for each processor.

Efficient job allocation is crucial for minimizing the longest processing time among all processors.

2. Understand the Algorithms

Part of the task for this assignment is to reimplement these algorithms, so it is important to ensure that you understand how they work and what they are supposed to do.